

Introduction to Linux

UNIX, Linux and GPL

Two important developments in modern computing occurred in 1969 at Bell Labs, a part of AT&T. A team developed the UNIX operating system, and the C programming language. The two combined are the foundations of modern operating systems. Unix was designed in a way to be implemented both on servers, as well as computers meant for individual use. The single most important contribution of UNIX, was to create standards on which independent software and hardware manufacturing companies could work on. This basically meant that a software programmer in one country knew his program would work on hardware from another country. UNIX was very influential, and widely used because of the complete documentation provided with the operating system.

Another important aspect of UNIX was the way it allowed computers to interface with each other. UNIX allowed tasks to be shared over a network, and thus lay the foundation of the Internet. Before UNIX, most operating systems were written in assembly level languages. An assembly level language is specific to the hardware for which it is written. If say, a game is written in an assembly language, then the game will run only on the hardware for which it is written, and not run on any other hardware. This means that any application written in an assembly language is not portable. The same holds true for operating systems as well.

UNIX was the first operating system to be written in the C programming language — a high-level programming language — which made UNIX portable. For the first time, a single operating system could run on different hardware, independent of the architecture of the computer system. Strictly speaking, an operating system written in a higher-level programming language compromises on a lot of benefits of operating systems programmed on assembly level programming languages, particularly noticeable in computing speed. Because of the way computers have grown though, this difference is no longer important, and C itself is considered a low level programming language because of the ease at which it can operate with a wide range of hardware.

UNIX was instantly picked up and widely used, spawning a lot of clone and derivative works. The derivatives include Sun OS, Solaris and BSD (Berkely Software Distribution). All of which are